

```
import org.springframework.beans.
factory.annotation.Autowired;
import org.springframework.stereotype.
Service;
import com.springai.image.
TextExtractionRequest;
import com.springai.image.
TextExtractionResponse;
import com.springai.image.ImageAI;
@Service
public class ChequeProcessingService {
    @Autowired
    private ImageAI imageAI;
    public String extractChequeText(byte[]
chequeImage) {
        TextExtractionRequest request =
new TextExtractionRequest(chequeImage);
        TextExtractionResponse response =
imageAI.extractText(request);
        return response.
getExtractedText();
    }
}
```

This example code creates a service that extracts text from an image of a cheque in a bank using the ImageAI component in the Spring AI framework. This extracted text can be used for cheque processing and database updates for transactions.

Audio transcription models

Audio transcription models convert spoken language into text. This capability is essential for creating accessible content and automating documentation processes. In healthcare, for instance, Spring AI can be leveraged to transcribe doctor-patient conversations, ensuring accurate record-keeping and enhancing patient care.

Here is an example code using AudioAI API in the Spring AI framework to transcribe audio to text (from a streaming byte array to text string):

```
import org.springframework.beans.
factory.annotation.Autowired;
import org.springframework.stereotype.
Service;
import com.springai.audio.
TranscriptionRequest;
```

```
import com.springai.audio.
TranscriptionResponse;
import com.springai.audio.AudioAI;
@Service
public class AudioTranscriptionService {
    @Autowired
    private AudioAI audioAI;
    public String transcribeAudio(byte[]
audioFile) {
        TranscriptionRequest request = new
TranscriptionRequest(audioFile);
        TranscriptionResponse response =
audioAI.transcribe(request);
        return response.getTranscription();
    }
}
```

Text to speech models

Text to speech models convert written text into spoken words, providing a means to create auditory content. This is especially beneficial in the wealth management industry, where financial advisors can use these to generate voice-based reports for clients, making complex financial information more accessible and comprehensible.

Here is an example code to create a service that converts the input text to speech using the SpeechAI component of the Spring AI framework.

```
import org.springframework.beans.
factory.annotation.Autowired;
import org.springframework.stereotype.
Service;
import com.springai.speech.
TextToSpeechRequest;
import com.springai.speech.
TextToSpeechResponse;
import com.springai.speech.SpeechAI;
@Service
public class TextToSpeechService {
    @Autowired
    private SpeechAI speechAI;
    public byte[]
convertTextToSpeech(String text) {
        TextToSpeechRequest request = new
TextToSpeechRequest(text);
        TextToSpeechResponse response =
speechAI.convert(request);
        return response.getSpeechAudio();
    }
}
```

```
}
```

Moderation models

Moderation models help ensure that content adheres to community guidelines by identifying and filtering inappropriate material. In a social media platform within the retail industry, Spring AI can be employed to automatically moderate user-generated content, maintaining a safe and friendly environment for all users. The code example given below uses ModerationAI of the Spring AI framework to create a service that checks whether the given advertisement text is appropriate for children.

```
import org.springframework.beans.
factory.annotation.Autowired;
import org.springframework.stereotype.
Service;
import com.springai.moderation.
ModerationRequest;
import com.springai.moderation.
ModerationResponse;
import com.springai.moderation.
ModerationAI;
@Service
public class ContentModerationService {
    @Autowired
    private ModerationAI moderationAI;
    public boolean
isContentAppropriate(String
advertisementText) {
        ModerationRequest request = new
ModerationRequest(advertisementText);
        ModerationResponse response =
moderationAI.moderate(request);
        return response.isAppropriate();
    }
}
```

Advanced features of Spring AI

In addition to its portable service abstractions that help developers to start building agentic AI applications, Spring AI provides several advanced features that make it a great choice for building intelligent AI applications using the Java framework.